

Database Systems Lab

Flask

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Flask Routes

Routes define URL patterns and their corresponding functions:

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route('/')
6 def home():
7     return 'Hello, World!'
8
9 @app.route('/user/<name>')
10 def user(name):
11     return f'Hello, {name}!'
12
13 @app.route('/posts/<int:post_id>')
14 def post(post_id):
15     return f'Post ID: {post_id}'
16
17 @app.route('/login', methods=['GET', 'POST'])
18 def login():
19     if request.method == 'POST':
```

Flask Templates (Jinja2)

Templates allow dynamic HTML generation:

```
1 # In your Flask route
2 @app.route('/profile/<username>')
3 def profile(username):
4     user = get_user(username)
5     return render_template('profile.html', user=
        user)
```

Template file (templates/profile.html):

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>{{ user.name }}'s Profile</title>
5 </head>
6 <body>
7     <h1>Welcome, {{ user.name }}!</h1>
8     {% if user.posts %}
9         <ul>
10             {% for post in user.posts %}
11                 <li>{{ post.title }}</li>
```

Basic Flask Directory Structure

Recommended project organization:

```
1 app/  
2 |- app.py                # Entry point  
3 |- config.json           # Configuration  
4 |- requirements.txt      # Dependencies  
5 |- routes/              # Route definitions  
6 |- static/              # Frontend, static  
7 |   |- css/  
8 |   |- js/  
9 |   |- images/  
10 |- templates/           # Frontend, template  
11 |   |- base.html  
12 |   |- index.html  
13 |   |- profile.html  
14 |- services/            # Backend  
15 |- ...                  # Backend
```

Flask Session

Sessions store user data across requests. It is used for the authentication state (e.g., logged-in user ID). It is stored client-side in a securely signed cookie, so the client can't tamper with it.

```
1 from flask import Flask, session, request,
   redirect
2
3 app = Flask(__name__)
4
5 @app.route('/login', methods=['POST'])
6 def login():
7     username = request.form['username']
8     if validate_user(username):
9         session['user_id'] = get_user_id(
            username)
10        session['username'] = username
11        return redirect('/dashboard')
12    return 'Invalid credentials'
13
14 @app.route('/dashboard')
15 def dashboard():
```

Flask g Object

The g object stores data during a single request. It is used for caching and maintaining database connections.

```
1 from flask import Flask, g, request
2
3 app = Flask(__name__)
4
5 @app.before_request
6 def load_user():
7     user_id = session.get('user_id')
8     if user_id:
9         g.user = get_user_by_id(user_id)
10    else:
11        g.user = None
12
13 @app.route('/profile')
14 def profile():
15     if g.user is None:
16         return redirect('/login')
17     return render_template('profile.html', user=
        g.user)
```

Flask App Object

The Flask app object is the core of your application:

```
1 from flask import Flask
2
3 # Create app instance
4 app = Flask(__name__)
5
6 # Configuration
7 app.config['SECRET_KEY'] = 'your-secret-key'
8 app.config['DATABASE_URL'] = 'sqlite:///app.db'
9
10 # Register blueprints
11 from routes.auth import auth_bp
12 from routes.main import main_bp
13 app.register_blueprint(auth_bp, url_prefix='/
    auth')
14 app.register_blueprint(main_bp)
15
16 # Error handlers
17 @app.errorhandler(404)
18 def not_found(error):
```

Conclusion

Thank you for your attention!